Dual Exposure Stereo for Extended Dynamic Range 3D Imaging

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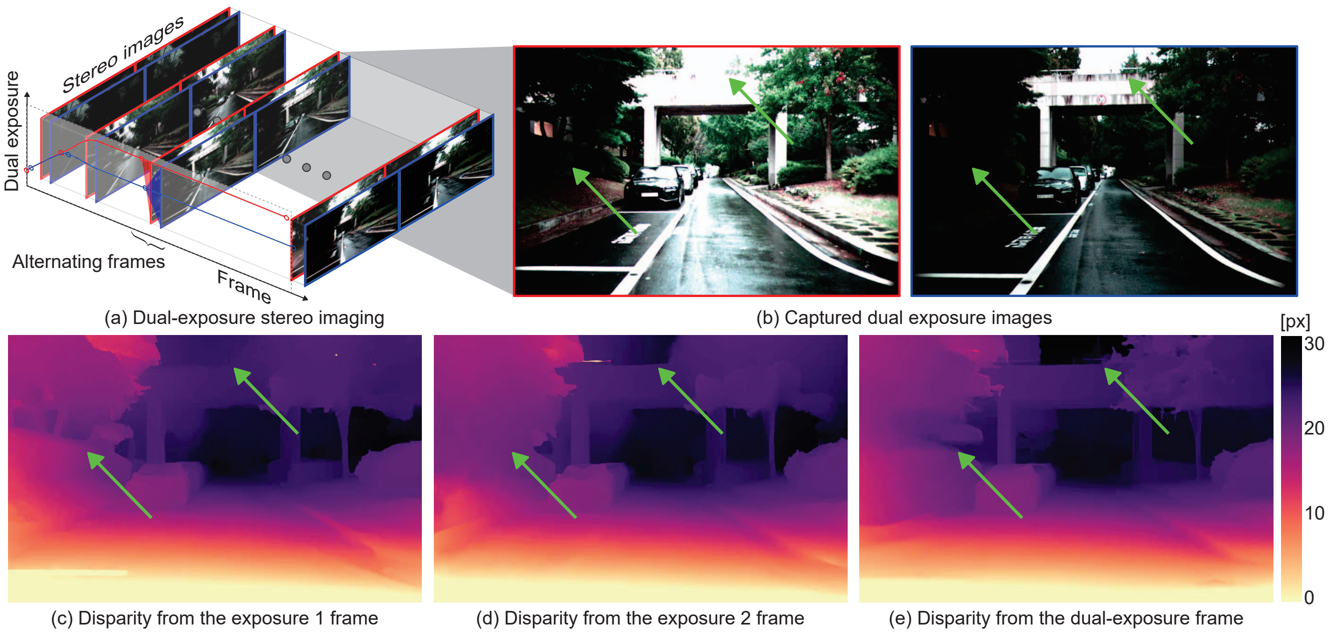


Figure 1. We introduce dual-exposure stereo, a method for extended dynamic range (DR) 3D imaging. (a) We control the dual exposures synchronously set for the stereo camera to expand the effective DR of 3D imaging. From (b) the captured dual-exposure stereo images, we estimate (e) a disparity map that preserves details in both the under- and over-exposed images of the dual-exposure pair, which cannot be faithfully reconstructed in (c)&(d) one-exposure results.

Abstract

Achieving robust stereo 3D imaging under diverse illumination conditions is challenging due to the limited dynamic range of conventional cameras, causing existing stereo depth estimation methods to suffer from under- or over-exposed images. In this paper, we propose dual-exposure stereo that combines auto-exposure control and dual-exposure bracketing to achieve stereo 3D imaging with extended dynamic range. Specifically, we capture stereo image pairs with alternating dual exposures, which automatically adapt to scene illumination and effectively distribute the scene dynamic range across the dual-exposure frames. We then estimate stereo depth from these dual-exposure stereo images by compensating for motion between consecutive frames. To validate our approach, we develop a robotic vision system, acquire real-world HDR stereo video datasets, and generate additional synthetic datasets. Experimental results demonstrate that our method outperforms existing exposure control methods.

1. Introduction

Robust 3D imaging is critical for autonomous systems, such as robots and self-driving vehicles, which depend on depth perception to navigate and interact with their environments. Stereo imaging is a popular 3D imaging technique that estimates depth from disparity by matching corresponding pixels in images captured by two cameras. Recent advancements in neural networks have significantly improved stereo disparity estimation, making stereo imaging a practical and cost-effective solution [27].

However, achieving robust 3D imaging with stereo cameras remains challenging, especially in real-world scenes that exhibit lighting conditions with ultra-wide dynamic ranges (DRs). Conventional cameras have limited DR capabilities [36], so in scenes with extremely wide DRs, bright regions may become overexposed while dark regions are underexposed, leading to suboptimal disparity estimation. Existing auto exposure control (AEC) methods adjust camera exposure to capture the scene DR, however they do not expand the camera's native DR, as each stereo frame is often processed individually [8]. Exposure bracketing techniques capture multiple images with different exposures to expand the effective DR through multi-exposure image processing [9, 28], however these often rely on predefined exposures that do not adapt to the scene DR, increasing capture time and computational overhead.

In this paper, we introduce dual-exposure stereo, a method for extended dynamic range 3D imaging (Figure 1). In alternating frames, we capture stereo images with dual exposures, producing two pairs of stereo images where each pair is captured at different exposure settings in successive frames. By combining AEC and exposure bracketing, we dynamically adjust the dual exposures: when the scene DR exceeds the camera's native DR, the dual exposures diverge to capture bright and dark regions across two successive frames. When the scene DR is within the camera's DR, the exposures converge to capture the full scene DR within the camera's native DR. Each stereo image captured under dual exposures retains the same DR as the camera, however different exposure settings enable coverage of bright and dark regions. To leverage these images, we develop a dual-exposure depth estimation method that fuses dual-exposure features in a motion-aware manner across alternating frames. Our approach effectively extends the DR for 3D imaging, regardless of the original bit depth of the cameras.

To validate our method, we design a robot-vision system equipped with stereo cameras and a LiDAR sensor. Using this setup, we collect a dataset of stereo videos and LiDAR point clouds across indoor and outdoor environments with various lighting conditions. We also generate synthetic datasets with dense ground-truth depth maps. Our experiments demonstrate that the proposed method outperforms previous exposure control methods, enabling depth estimation in scenes with a wide range of DRs. Code and datasets will be made publicly available.

In this paper, we make the following contributions:

- We introduce dual-exposure stereo for extended dynamic range 3D imaging, developing an automatic dual-exposure control method that combines conventional AEC and exposure bracketing. Our dual-exposure disparity estimation method then utilizes dual-exposure stereo images to increase effective camera DR for robust 3D imaging.
- We develop a robot-vision system with stereo cameras and a LiDAR sensor mounted on a wheeled robot, collecting a real-world stereo video dataset and rendering a synthetic dataset with dense ground truth.
- We validate our method on both synthetic and real-world datasets, demonstrating improved performance over existing exposure control methods.

2. Related Work

HDR 3D Imaging Active imaging systems with engineered illumination have enabled 3D imaging under high-dynamic-range (HDR) environments. Examples are synchronized projector-camera systems [33, 34] and time-of-flight cameras [7, 41]. Without additional illumination modules, passive 3D imaging systems exploit unconventional sensors for HDR imaging, such as single-photon avalanche diodes [17], quanta image sensors [11, 13], event cameras [46, 55, 59], and modulo cameras [57, 58]. Using conventional cameras, capturing scenes with multiple exposures is a standard approach for HDR 3D imaging [4, 5, 23]. However, these methods rely on predetermined exposure settings, which cannot adapt to changing lighting conditions, leading to loss of detail in overexposed or underexposed regions and unnecessarily long acquisition times when scene DR is low. Our method automatically adjusts dual exposure for stereo cameras, improving performance for varying-DR scenes by expanding effective camera DR for 3D imaging.

AEC and Exposure Bracketing Single-camera AEC has been extensively studied using histogram analysis [2, 39, 44], model-predictive correction [35, 42, 43, 45], entropy analysis [25, 31, 54], and semantic analysis [32, 52, 53]. Extending AEC to stereo cameras, there are two common approaches. The first is to control exposure for each camera individually. However, this deteriorates stereo correspondence due to stereoscopic intensity inconsistency [21, 56]. The second approach is to use a synchronized AEC for stereo cameras, maintaining intensity consistency [21, 40]. However, existing methods in this category struggle when scene DR is larger than camera DR, leading to overexposed or underexposed regions. Exposure bracketing alternates multiple exposures and processes the multi-exposure images to capture a broader DR than the original camera DR [14, 15, 30, 32, 37, 47–49]. Most existing methods use a single camera [16, 50] and rely on predetermined long and short exposure settings [3, 6, 12, 18–20, 26], which limits capture efficiency when scene DR fluctuates. Our method combines the principles of AEC and exposure bracketing. We control the dual exposures of stereo cameras, thus maintaining intensity consistency between stereo images as well as expanding effective DR for 3D imaging.

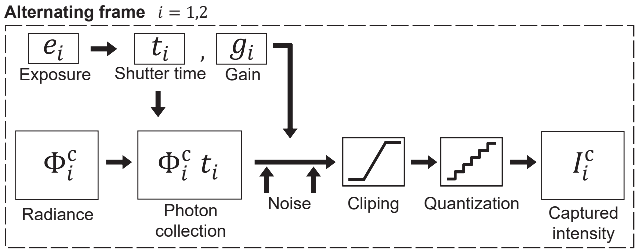


Figure 2. **Image Formation.** For alternating frame $i = \{1, 2\}$, scene radiance Φ_i, and exposure e_i, we simulate the captured intensity I_i^c for the camera $c \in \{\text{left}, \text{right}\}$. We consider photon collection, pre- and post-gain noise, clipping, and quantization.

Stereo Depth Estimation Estimating depth from stereo disparity has been studied for decades [38]. Recent neural-network solutions have significantly improved stereo matching by using deep feature extraction and cost volume construction [22, 24]. These models iteratively refine disparity maps using 3D convolutions or recurrent units. However, they fail when scene DR is wider than camera DR, because of under- and over-exposed regions. Our disparity estimation method addresses this problem by fusing the dual-exposure images while compensating the motion between consecutive frames.

3. Image Formation

We introduce the dual-exposure image formation model for stereo cameras. We denote the dual exposure as e_i, where $i \in \{1, 2\}$ is an alternating frame index. We convert the exposure e_i to shutter time $t_i = e_i/g_i$ and gain $g_i = \max(1, e_i/t_{\text{max}})$, where t_{max} is the maximum shutter time, allocating shutter time as long as possible capped by the maximum value to reduce image noise by a high gain. Given the shutter time t_i, gain g_i, and the incident scene radiance Φ_i, we model the intensity I_i^c captured by the stereo camera $c \in \{\text{left}, \text{right}\}$ at the frame i as:

$$I_i^c(p_i^c) = \text{quant}\left(\text{clip}(g_i(\Phi_i t_i + n_i^{\text{pre}}) + n_i^{\text{post}})\right), \quad (1)$$

where p_i^c is a camera pixel. The noise terms n_i^{pre} and n_i^{post} are the pre-gain and post-gain noise, sampled from zero-mean Gaussian distributions with standard deviations σ_{pre} and σ_{post}, respectively: $n_i^{\text{pre}} \sim \mathcal{N}(0, \sigma_{\text{pre}})$, $n_i^{\text{post}} \sim \mathcal{N}(0, \sigma_{\text{post}})$. The function $\text{clip}(\cdot)$ limits intensity values by the camera DR, and $\text{quant}(\cdot)$ quantizes the intensity to integer values. The overall procedure of the image formation is shown in Figure 2.

4. Auto Dual Exposure Control

Our ADEC method uses the left-camera images $\{I_1^{\text{left}}, I_2^{\text{left}}\}$ captured by the dual exposure $\{e_1, e_2\}$ to estimate the next dual exposure $\{\hat{e}_1, \hat{e}_2\}$. Pseudo code of our ADEC method is shown in Algorithm 1. Figure 3 shows an example scenario of applying the ADEC method.

Algorithm 1 Pseudocode for ADEC.

Require: Dual exposure values e_1, e_2 and corresponding captured images $I_1^{\text{left}}, I_2^{\text{left}}$
Ensure: Next dual exposure $\hat{e}_1, \hat{e}_2$

```
// Compute metrics
for each frame i ∈ {1, 2} do
    h_i ← ExtractHistogram(I_i^left)
    S_i ← Skewness(h_i)
    L_i, H_i ← ExtremePixelRatios(h_i)
end for
// Adjust dual exposure
if Scene DR > camera DR: L_i > τ_h and H_i > τ_h for any i then
    if Dual exposure gap is low: Δe = |e_1 − e_2| ≤ τ_Δe then
        // Diverge dual exposure
        ê_1, ê_2 ← DivergeDualExposure(e_1, e_2, L_1, L_2, H_1, H_2)
    end if
else
    // Scene DR > camera DR or scene DR is uncertain
    for each frame i ∈ {1, 2} do
        // Adjust dual exposure towards zeroing the skewness
        ê_i ← MakeSkewnessZero(e_i, S_i)
    end for
end if
```

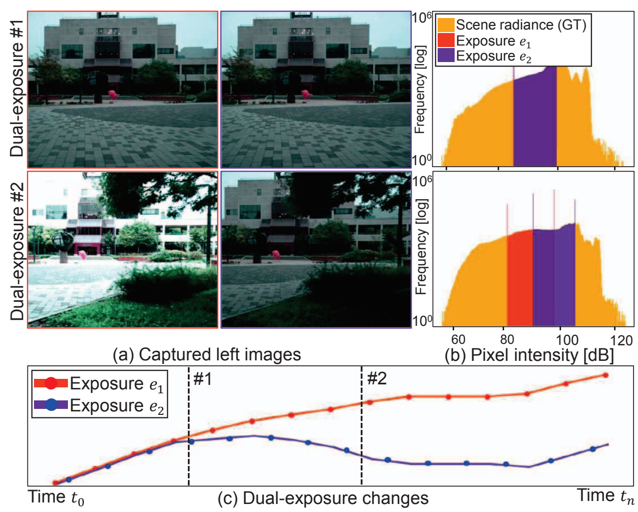


Figure 3. **ADEC Method Exposure Selection.** The initial dual exposures are set to be the same in this example. If the scene DR is estimated as wider than the camera DR, the dual exposures diverge to capture both dark and bright regions in alternating frames.

Metric Our ADEC method uses statistical metrics to control next-frame dual exposures. Specifically, for each frame $i \in \{1, 2\}$, we compute the intensity histogram h_i of the image I_i^{left}, and calculate the histogram skewness S_i, describing whether the histogram is skewed towards low intensity (negative skewness) or high intensity (positive skewness) as

$$S_i = \sum_{j=0}^{K} \left(\frac{j - K/2}{K/2}\right)^3 \frac{h_i(j)}{N}, \quad (2)$$

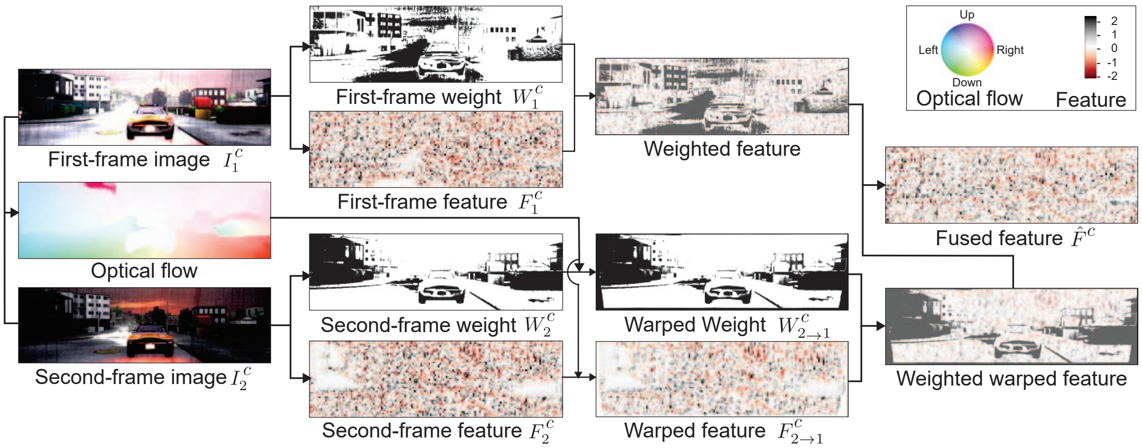


Figure 4. **Dual Exposure Feature Fusion.** For each camera c, we extract features F_1^c and F_2^c, optical flow f^c, and weight maps W_1^c and W_2^c of the dual-exposure images I_1^c and I_2^c. The second-frame features and weight map are warped to the first-frame view, and fused to the final feature $\hat{F}^c$ with the weighted summation, encoding dual-exposure information.

where K is the maximum detectable intensity of the camera depending on its bit depth, N is the number of pixels, and $h_i(j)$ is the frequency of intensity j.

We then compute the ratios of under- and over-exposed pixels, denoted as L_i and H_i, representing the proportions of pixels near the minimum and maximum intensities

$$L_i = \sum_{j=0}^{T_{\text{low}}} \frac{h_i(j)}{N}, \quad H_i = \sum_{j=T_{\text{high}}}^{K} \frac{h_i(j)}{N}, \tag{3}$$

where $T_{\text{low}} = \lfloor K \times 0.05 \rfloor$ and $T_{\text{high}} = \lfloor K \times 0.95 \rfloor$ are clamping thresholds.

Below, we use the skewness S_i, extreme-valued pixel ratios L_i, H_i to determine the next-frame dual exposures.

Diverging Dual Exposure For scenes whose DR exceeds the camera native DR, we diverge the dual exposure to capture a broader DR across the alternating frames. Such cases are identified with the conditions $L_i > \tau_h$ and $H_i > \tau_h$ for at least one frame i, where $\tau_h = 0.05$. We diverge the dual exposure with magnitudes proportional to L_i and H_i as

$$\begin{aligned} &\hat{e}_1 = e_1 + \alpha L_1, \quad \hat{e}_2 = e_2 - \alpha H_2, \quad \text{if} \quad e_1 > e_2 \\ &\hat{e}_1 = e_1 - \alpha H_1, \quad \hat{e}_2 = e_2 + \alpha L_2, \quad \text{otherwise}, \end{aligned} \tag{4}$$

where $\alpha = 0.5$ is a constant controlling the divergence step.

To prevent excessive divergence that could lead to unstable stereo imaging, we limit the exposure difference $\Delta e = |e_1 - e_2|$. If Δe exceeds a threshold $\tau_{\Delta e} = 2.5$, no further divergence is applied.

Towards Zeroing Skewness When the scene DR is uncertain or lower than the camera DR, indicated by $L_i < \tau_h$ and $H_i < \tau_h$ for both frames $i \in \{1, 2\}$, we adjust the exposures towards zeroing the skewness S_i. This makes the intensity histogram to be balanced, capturing the scene DR:

$$\hat{e}_i = e_i - \alpha \times S_i. \tag{5}$$

This process makes the dual exposure converge to a similar value, to fully cover the scene DR in a balanced manner, if the scene DR is lower than camera DR. For scenes with uncertain DR compared to the camera DR, this method moves the dual exposure to be with zero skewed, facilitating better identification of the scene DR.

5. Dual-exposure Stereo Disparity Estimation

We introduce our disparity estimation method for dual-exposure stereo images: I_1^{left}, I_1^{right} of the first frame, and I_2^{left}, I_2^{right} of the second frame.

Motion Estimation Between alternating frames $i \in \{1, 2\}$, the locations of corresponding pixels can move for dynamic movements of objects or cameras, modeled as the optical flow f^c:

$$p_1^c = p_2^c + f^c(p_2^c). \tag{6}$$

To account for such motion, we estimate the optical flow f^{left} between I_1^{left} and I_2^{left}, and f^{right} between I_1^{right} and I_2^{right} using a pretrained optical flow network robust to intensity difference between images [29]. Note that the dual-exposure images I_1^c and I_2^c have overlapped contents as we do not allow for extremely-diverged dual exposures as described in Section 4.

The estimated optical flow f^c allows us to define a warping function that transforms data $\mathcal{X}_2^c$ from the second frame ($i = 2$) to the first frame ($i = 1$):

$$X_{2\to1}^c = \text{warp}(X_2^c, f^c), \tag{7}$$

where $X_{2\to1}^c$ is the warped data. The warping function entails resizing the optical flow field to the corresponding resolution of the input data X_2^c.

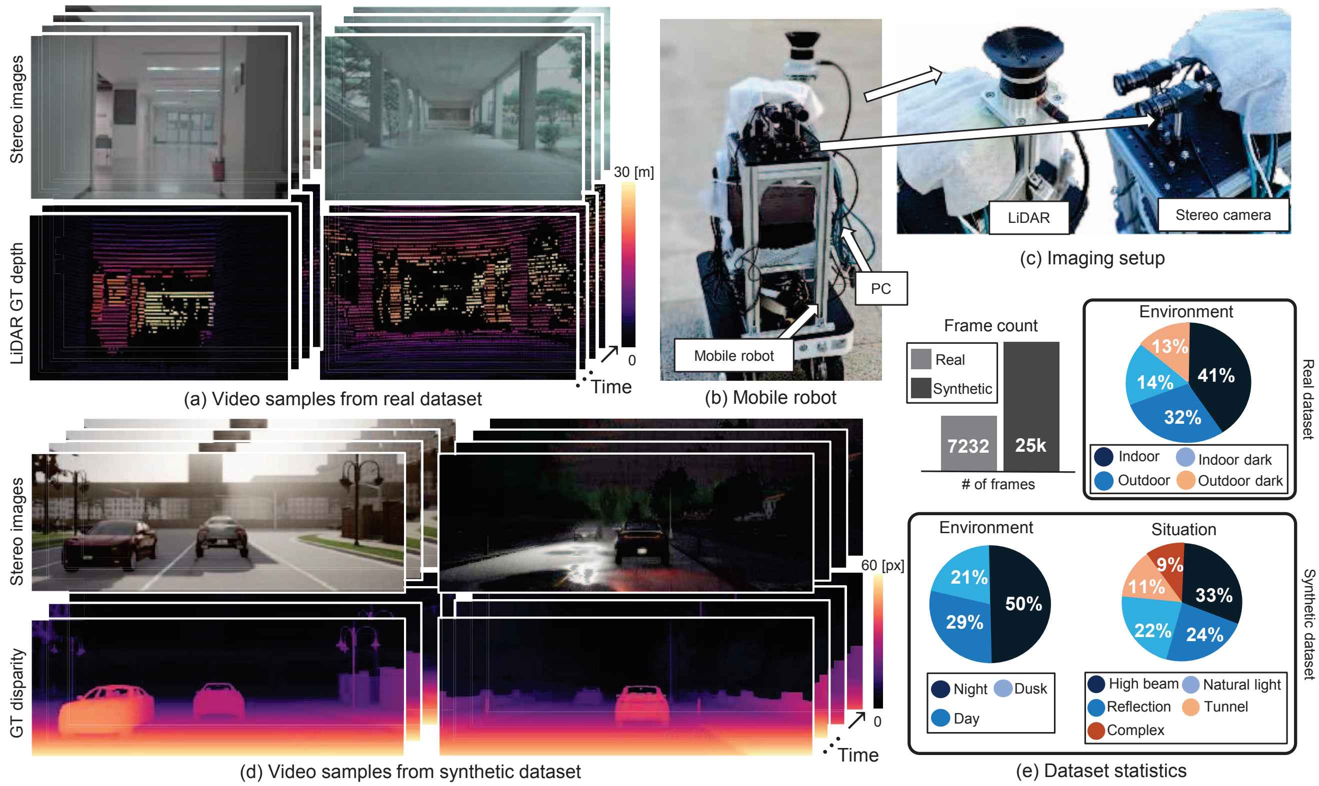


Figure 5. **Stereo Video Datasets.** (a) Samples of our real-world dataset: sequence of stereo images and corresponding LiDAR point cloud. (b)&(c) Our imaging setup (c) mounted on a mobile robot (b), equipped with a LiDAR sensor and stereo cameras. (d) Samples of our synthetic dataset: stereo images and ground-truth disparity maps. (e) Statistics of the synthetic dataset about time, location, and scenarios.

Dual-exposure Feature Fusion To exploit dual-exposure images $\{I_1^c, I_2^c\}$ with potentially different exposures, we develop a dual-exposure feature fusion method. Figure 4 shows the overview of our dual-exposure feature fusion. We extract features F_i^c from images I_i^c using a pretrained feature extractor $\mathrm{FE}(\cdot)$ [24]:

$$F_i^c = \mathrm{FE}(I_i^c). \tag{8}$$

We then warp the second-frame feature to the first frame using the warping function based on the estimated optical flow, ensuring that feature from the second frame becomes spatially aligned with the first-frame feature:

$$F_{2\to1}^c = \mathrm{warp}(F_2^c, f^c). \tag{9}$$

As we now have the dual exposure features $F_{2\to1}^c$ and F_1^c spatially aligned, we fuse the two features using the weighted sum:

$$\hat{F}^c = \frac{W_1^c \cdot F_1^c + W_{2\to1}^c \cdot F_{2\to1}^c}{W_1^c + W_{2\to1}^c + \epsilon}, \tag{10}$$

where $\hat{F}^c$ is the fused feature, ϵ is a small constant to avoid division by zero.

We define the weight map W_i^c using an intensity-based trapezoidal function [19] to exploit well-exposed pixel intensity:

$$W_i^c(p_i^c) = \begin{cases} \frac{I_i^c(p_i^c)}{\alpha} & \text{if } I_i^c(p_i^c) < \alpha, \\ 1 & \text{if } \alpha \le I_i^c(p_i^c) \le \beta \\ 1 - \frac{1}{1-\beta}\left(I_i^c(p_i^c) - \beta\right) & \text{if } I_i^c(p_i^c) > \beta, \end{cases} \tag{11}$$

where $\alpha = 0.02$ and $\beta = 0.98$ are the thresholds. The weight maps W_1^c and $W_{2\to1}^c$ are computed for the first frame and the second frame followed by being warped to the first-frame using the estimated optical flow.

We apply the dual-exposure feature fusion of Equation (10), obtaining the fused feature maps $\hat{F}^{\text{left}}$ and $\hat{F}^{\text{right}}$.

Stereo Disparity Estimation Using the fused feature maps, we construct correlation volumes C as

$$C(x, y, d) = \hat{F}^{\text{left}}(x, y) \cdot \hat{F}^{\text{right}}(x + d, y), \tag{12}$$

where x, y are the pixel coordinates and d is the disparity. Our dual-exposure feature fusion encodes both dark and bright features of dual-exposure stereo images in the correlation volume, allowing for effectively extended DR for 3D imaging. We also apply a multi-scale feature fusion approach for robustness. We then estimate a disparity map from the correlation volume using a disparity estima-

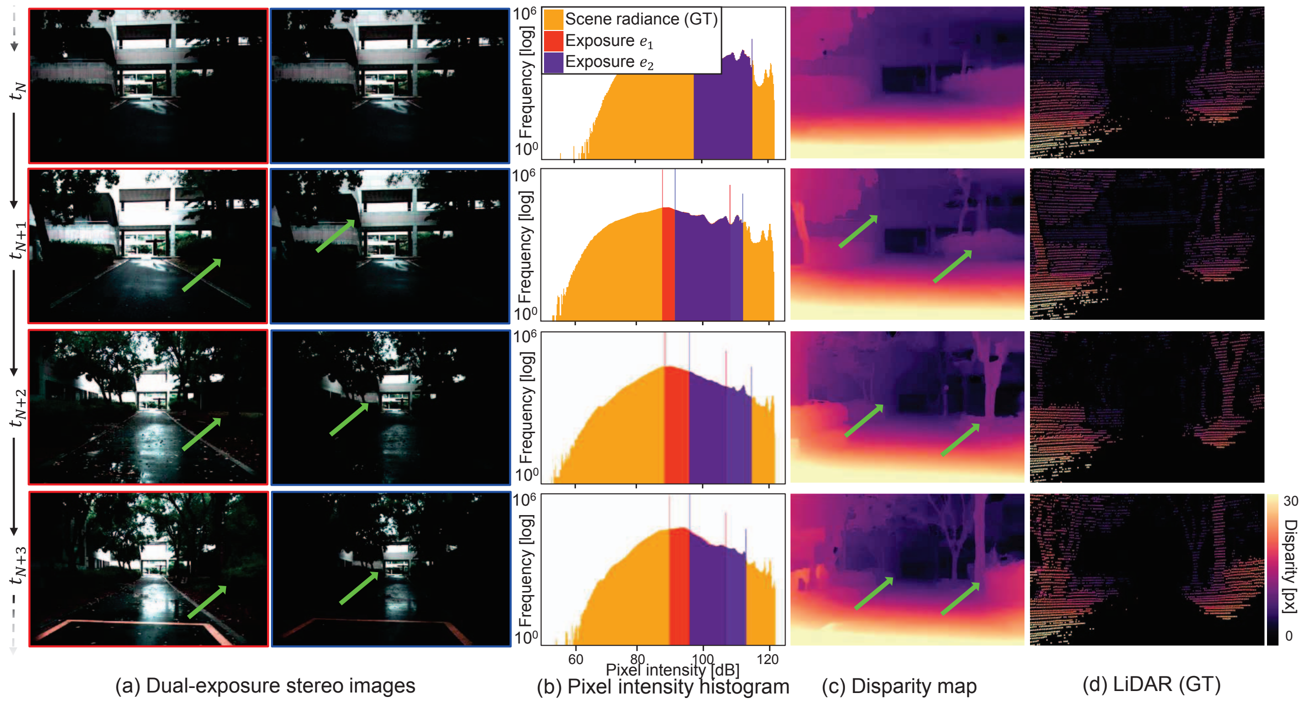


Figure 6. **Extended-DR 3D imaging.** Our ADEC controls the dual exposures for extended-DR depth estimation. The dual-exposure depth estimation module obtains depth both for bright and dark regions which cannot be captured by only one image.

tion network [24]. We finetune the network on our synthetic dataset (Section 6.2).

6. Stereo Video Datasets

To validate our method, we introduce two stereo video datasets: one for real-world scenes and the other for synthetic scenes. Figure 5 visualizes samples from our datasets.

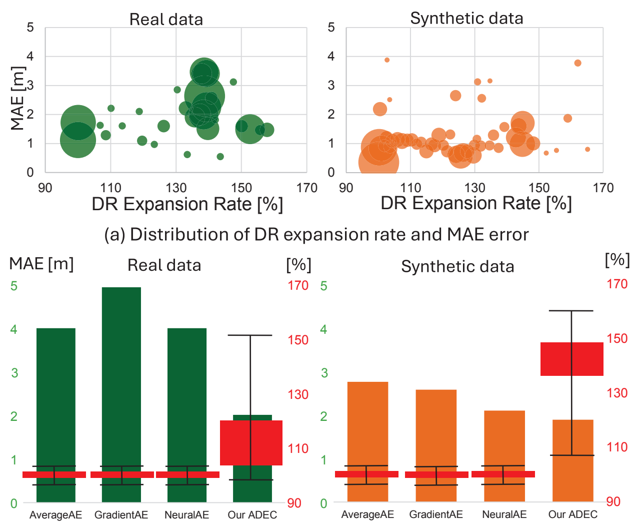
6.1. Real-world Dataset

Prototype System We developed an imaging system consisting of stereo RGB cameras (LUCID Triton 5.4MP) and a LiDAR sensor (Ouster OS-1). The cameras capture linear stereo images with 24-bit depth, which are then compressed to simulate 8-bit images using Equation (1) to evaluate whether our method can expand effective DR for 3D imaging. Note that our method can be applied to cameras with any bit depth. We performed intrinsic and extrinsic calibration of the stereo camera system using a chessboard-based method. Subsequently, we estimated the extrinsic transformation between the LiDAR and the left camera through ICP alignment process. LiDAR point clouds are projected onto the left-camera view, providing pseudo ground-truth sparse depth maps. The stereo images and LiDAR depth maps are time-synchronized. We mount the imaging system on a wheeled robot (AgileX Ranger-Mini 2.0), as shown in Figure 5(c), enabling both indoor and outdoor captures. We configured a stereo camera system with a 110 mm baseline to enable effective depth estimation in both indoor and outdoor environments. The depth measurement range of our system extends up to 60 meters. For system details, we refer to the Supplementary Document.

Captured Dataset Our real-world dataset encompasses a broad range of environments and lighting conditions, thus appropriate for evaluating our method. The dataset includes 33 scenes and 7432 frames, with a resolution of 1440×928 pixels. The dataset is balanced with 41% of frames captured in indoor scenes, 32% in outdoor scenes, 14% in indoor low-light scenes, and 13% in outdoor low-light scenes. Further details are provided in the supplementary material.

6.2. Synthetic Dataset

We use the CARLA simulator [10] to generate a synthetic dataset for diverse automotive scenarios with varying lighting conditions. We render synchronized stereo images with 32 bit depth, which is used to simulate 8-bit images using Equation (1). We also render ground-truth dense disparity maps. Our synthetic dataset comprises 1,000 training videos and 200 testing videos. Training videos consist of 20 frames each, and testing videos contain 100 frames each. We simulate various lighting conditions (day, dusk, and night), as shown in Figure 5, which vary within each video, introducing abrupt changes in dynamic range due to environmental effects such as high beams at night, intense reflections, and sunlight emerging from tunnels.



(b) Distribution of DR expansion rate and MAE error across methods

Figure 7. **Disparity accuracy vs. DR expansion rate.** (a) We demonstrate effective DR expansion up to 160% without significant performance drop. (b) Other AEC methods [1, 32, 40] cannot expand DR resulting in large error. The red boxes represent the distribution of DR expansion rates (median as dashed line).

7. Experiments

Extended-DR 3D Imaging We leverage complementary information of two-frame stereo images captured with dual exposure, which is estimated by our ADEC module. By fusing dual-exposure features, we obtain accurate disparity map that preserves details across a wider DR than the native camera DR. Figure 6 reports the estimated disparity maps for input dual-exposure stereo images. Our method enables expanding effective DR for disparity estimation with both details of dark and bright regions, which cannot be solely captured with the camera DR.

We assess the trade-off between the depth accuracy and the DR expansion rate. Compared to the maximum DRs of 42dB corresponding to 8-bit depth, we calculate the effectively-enhanced DR covered by the dual-exposure frames in our method. Figure 7 shows that our method retains high depth accuracy for the DR expansion rate of 160%, which demonstrates the effectiveness of our method. In contrast, depth estimation using other state-of-the-art AEC methods [1, 32, 40] cannot expand the DR, resulting in large error in depth reconstruction.

Comparison We compare our method with three state-of-the-art AEC methods that control and process single exposure [1, 32, 40]. For a fair comparison, we finetune both the AEC module and the stereo network end-to-end. We use the RAFT-Stereo model [24] for their depth estimation. Note that ours is the only method that combines the principles of AEC and exposure bracketing, which is exploited by our dual-exposure disparity estimation module. Figures 8 and 9 show the qualitative results on a synthetic scene and a real-world scene, demonstrating that our method only recovers details in both dark and bright regions by effectively expanding DR for 3D imaging. Table 1 confirms that our method also quantitatively outperforms the other AEC methods for accurate 3D imaging on both synthetic and real-world datasets. We also compare the speeds of AEC methods and our ADEC method considering its real-time use cases, which is an important factor for any exposure control methods. Our ADEC can run at more than 120 FPS, supporting real-time applications, while the neu-

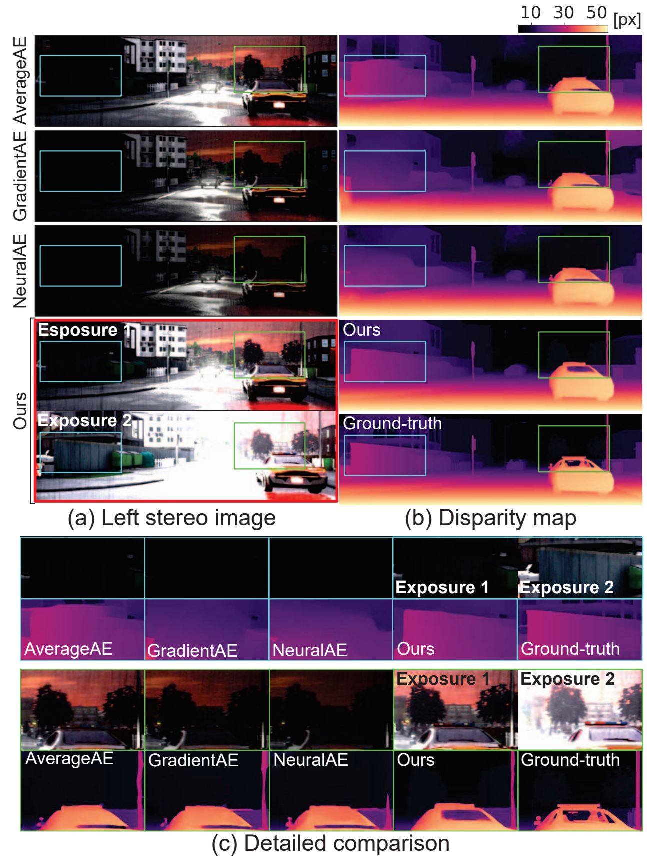


Figure 8. **Disparity accuracy using our ADEC compared with other AEC methods.** Our method outperforms the other AEC methods: AverageAE [1], GradientAE [40], and NeuralAE [32].

	AverageAE [1]	GradientAE [40]	NeuralAE [32]	ADEC (ours)
Synthetic Data Disp. MAE [px] ↓	2.2958	2.4680	<u>1.7198</u>	**1.3837**
Real Data Depth MAE [m] ↓	2.7089	2.6677	<u>1.7565</u>	**1.6518**
FPS↑	**616.27**	42.10	0.25	<u>124.58</u>

Table 1. **3D imaging accuracy and FPS comparison of our ADEC against other methods.** Our method outperforms the other AEC methods in terms of disparity and depth MAE on synthetic and real datasets, respectively. We also compare FPS. Best numbers are in **bold** and the second best are in <u>underline</u>.

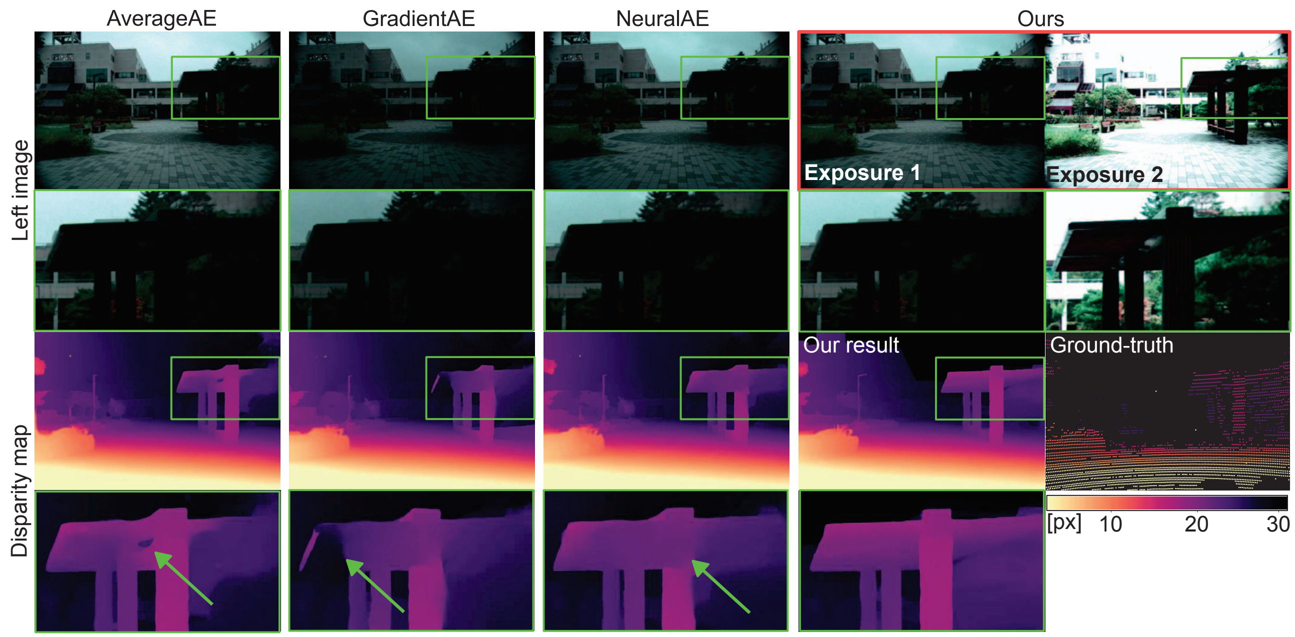


Figure 9. **Disparity-estimation results using our ADEC compared with other AEC methods.** Our ADEC method outperforms the other AEC methods for subsequent extended-DR depth estimation: AverageAE [1], GradientAE [40], and NeuralAE [32].

Study	Configuration: Backbone	Configuration: AEC	Motion compensation	Disp. MAE [px] ↓
Network backbone	[24]	[1]	✓	2.58
	[24]	[40]	✓	2.52
	[24]	[32]	✓	1.92
	[24]	Ours	✓	**1.76**
	[51]	[1]	✓	2.18
	[51]	[40]	✓	1.79
	[51]	[32]	✓	1.67
	[51]	Ours	✓	**1.46**
Fusion strategies	Exposure-level fusion		×	3.22
	Exposure-level fusion		✓	2.65
	Disparity-level fusion		×	5.03
	Disparity-level fusion		✓	3.75
	Feature-level fusion		×	4.02
	Feature-level fusion		✓	**1.82**
Module analysis	w/o ADEC		✓	6.28
	w/o weighted fusion		✓	3.39
	w/o motion compensation		×	8.37
	Full model		✓	**2.90**

Table 2. **Ablation study.**

ral AEC [32] fails to support real-time imaging. The FPS values in Table 1 are for the exposure control modules only.

Ablation Study Table 2 shows ablation study on stereo backbones, feature fusion strategies, motion compensation, and the role of each individual module. First, replacing the RAFT-Stereo backbone [24] with a more recent model [51], our ADEC outperforms other AEC methods consistently with the case of using RAFT-Stereo [24]. Second, we compare (1) fusing dual-exposure images before stereo matching, (2) merging separately estimated disparity maps, and (3) our intermediate feature fusion approach. The results show that our feature-level fusion provides the most accurate results. Third, we assess the impact of motion compensation by examining disparity estimation without optical flow alignment. The results show that removing motion compensation introduces significant depth error. Lastly, we ablate the three core components of our method: the ADEC module, weighted feature fusion, and motion compensation. The results show that each component contributes to improved disparity accuracy.

8. Conclusion

In this work, we introduce a dual exposure stereo for extended DR 3D imaging. We devise a ADEC module and dual-exposure depth estimation method, expanding the effective DR for robust 3D imaging. To validate this method, we report a stereo video dataset consisting of stereo videos and LiDAR pointclouds, collected by our robot vision system. We evaluate the effectiveness of our method, outperforming conventional AEC methods across all experimental settings we tested. As a method that can expand the DR of any stereo camera for 3D imaging, we hope the proposed approach can be a step to depth imaging in even more extreme lighting and enviromental conditions, including photon-starved captures in fog, rain, or snow.

Acknowledgments Seung-Hwan Baek was supported by Korea NRF (RS-2023-00211658, RS-2024-00438532, 2022R1A6A1A03052954), and Samsung Research Funding & Incubation Center of Samsung Electronics under Project Number SRFC-IT1801-52.

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